Topic 2 – Performance characteristics

2.1 Performance characteristics of materials

Spec: Conductivity, strength, elasticity, plasticity, malleability, ductility; Hardness, toughness, durability, biodegradability; Applied to woods, metals, polymers, smart and modern materials, papers, boards, textiles and composites

Key facts

• **Define toughness.** The ability to absorb impact energy without breaking or shattering.
• **Hardness vs toughness – give an example of a hard but brittle material.** Cast iron, glass, acrylic and high carbon steel (untempered) are hard but brittle.
• **Define malleability and give a highly malleable metal.** Ability to be hammered/pressed/rolled into shape without cracking – gold, copper, aluminium, lead.
• **Define ductility and give an example.** Ability to be drawn into wire without breaking – copper (electrical wire).
• **Elasticity vs plasticity?** Elastic: returns to shape when load removed. Plastic: permanent deformation without breaking.
• **What does the area under a stress–strain graph represent?** The energy absorbed before fracture – a measure of toughness.
• **Workshop test for hardness?** Centre punch / indentation test – compare size of dent; or scratch test (file). Industrial: Brinell, Vickers, Rockwell.
• **Workshop test for toughness?** Hit a clamped sample with a hammer and see if it bends or shatters. Industrial: Izod or Charpy pendulum impact test.
• **What is tensile testing?** Sample is pulled until it breaks in a tensometer; gives yield stress, UTS, elongation and stiffness.
• **Which materials are biodegradable?** Natural materials: wood, paper and board, cotton, wool, linen, natural rubber (slowly), bioplastics like PLA (industrial composting).
• **What is durability for a garden bench?** Resistance to rot, weathering, UV and wear so it keeps working and looking good for many years.

Characteristic	Meaning	High examples	Low examples
Conductivity (thermal/electrical)	Lets heat or electricity pass through	Copper, aluminium, silver	Polymers, wood, ceramics, foam, wool (insulators)
Strength	Resists a force without breaking – tensile, compressive, shear, torsion, bending	Steel, CFRP (tensile); concrete, cast iron (compressive)	Balsa, LDPE, paper
Elasticity	Returns to original shape when load removed	Rubber, spring steel, SMA (superelastic)	Clay, lead
Plasticity	Deforms permanently without breaking	Heated thermoplastics, clay, lead	Glass, cast iron
Malleability	Deforms in all directions under compression (hammering, rolling, pressing)	Gold, copper, aluminium, lead, mild steel	Cast iron, zinc (cold), glass
Ductility	Can be stretched/drawn into wire	Copper, gold, mild steel, aluminium	Cast iron, acrylic
Hardness	Resists scratching, wear, indentation	Tool steel, thermo-ceramics, glass, UF	Lead, LDPE, balsa
Toughness	Absorbs impact without fracture	Mild steel, ABS, polycarbonate, PP, rubber	Glass, cast iron, acrylic, ceramics
Durability	Withstands wear, weathering and deterioration over time	Stainless steel, HDPE, oak, GRP	Paper, untreated pine, unprotected mild steel
Biodegradability	Broken down naturally by microorganisms	Wood, paper and board, cotton, wool, PLA	Most polymers, metals, glass

Property	Workshop test	Industrial test
Hardness	Indent with a centre punch and hammer, or scratch with a file; compare marks	Brinell (steel ball), Vickers (diamond pyramid), Rockwell – measure indentation size/depth
Toughness	Hit a sample held in a vice with a hammer – does it bend or shatter?	Izod/Charpy pendulum impact test on a notched specimen
Tensile strength / elasticity	Hang weights from samples and measure deflection or extension	Tensometer pulls a sample to failure, plotting load against extension
Ductility / malleability	Bend a strip over 90° in a vice – cracks on the outside show low ductility, on the inside low malleability	Bend test, elongation % from tensile test
Conductivity	Measure resistance with a multimeter; time how long heat takes to travel along a bar	Four-point probe resistivity; thermal conductivity rig
Corrosion/durability	Leave samples outside or in salt water	Salt-spray cabinet, accelerated weathering (UV)

Key terms

Conductivity	How well a material allows heat (thermal) or electricity (electrical) to pass through it. Metals are good conductors; polymers, wood and ceramics are insulators.
Strength	The ability to withstand a force without breaking or permanently deforming. Types: tensile (pulling), compressive (squashing), shear (sliding), torsional (twisting), bending.
Elasticity	The ability to deform under a load and return to the original shape when the load is removed.
Plasticity	The ability to be permanently deformed (reshaped) without breaking, once the elastic limit is passed.
Malleability	The ability to be permanently deformed in all directions by hammering, pressing or rolling without cracking.
Ductility	The ability to be stretched (drawn) out into a long thin wire without breaking.
Hardness	Resistance to scratching, abrasion, wear, cutting and indentation.
Toughness	The ability to absorb energy/sudden impact without fracturing. The opposite of brittleness.
Durability	The ability to withstand wear, weathering, corrosion and deterioration over time.
Biodegradability	The ability to be broken down naturally by microorganisms (bacteria, fungi) into water, CO_2 and biomass.
Elastic limit / yield point	The stress beyond which a material no longer returns to its original shape and deforms plastically.
Young's modulus	Stiffness: stress ÷ strain in the elastic region. A steep line on a stress–strain graph means a stiff material.

Exam tip: Hard ≠ tough. Hardened high carbon steel and glass are hard but brittle. Use "toughness" only for resisting **impact**.